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AUGUST 1, 1933
SUPERSEDING
F.S. No. 424
July 9, 1926

FEDERAL SPECIFICATION

ROOF-COATING; ASPHALT, BRUSHING-CONSISTENCY

This specification was approved for promulgation by the Federal Specifications Board on August 1, 1933, for the use of the departments and independent establishments of the Government in the purchase of this commodity, and shall become mandatory for all departments and independent establishments of the Government not later than February 15, 1934. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE FEDERAL SPECIFICATIONS

A-1. There are no other Federal specifications applicable to this specification.

A-2. Any special requirements of the individual departments of the Government are noted under section H.

B. TYPE

B-1. Asphalt roof coating shall be of one type only.

C. MATERIAL

C-1. The coating shall consist of asphaltic materials, with or without fatty oils, thinned with a volatile solvent to a heavy brushing consistency, and mineral filler.

D. GENERAL REQUIREMENTS

D-1. *Basis of purchase.*—The material shall be purchased by weight or volume, as specified in the contract.

E. DETAIL REQUIREMENTS

E-1. *Condition in container.*—The material shall not liver in the container and after slight stirring shall be a smooth homogeneous mixture.

E-2. *Consistency.*—It shall be of a heavy brushing consistency to insure a film of appreciable thickness and yet not so stiff as to prevent easy application with a 3-knot roof brush at the rate of approximately 1 gallon per 50 square feet over asphalt roofing.

E-3. *Nonvolatile matter.*—The roof coating shall yield not less than 60 percent nonvolatile matter when 10 grams are heated in an oven at 105° C. (221° F.) for 24 hours.

E-4. *Mineral filler.*—The material shall contain not less than 5 percent and not more than 25 percent of a suitable mineral filler. The residue obtained from the ignition of the mineral filler shall be not less than 85 percent.

E-5. *Behavior at 60° C. (140 F).*—Test pieces prepared by brushing the coating on smooth-surfaced asphalt prepared roofing and metal at the rate of 1 gallon to 50 square feet, exposed for 60 minutes in a well-ventilated room and then in an oven at 60° C. (140° F.) for 5 hours shall show no blistering, and not more than one-fourth inch sagging or slipping, and no appreciable absorption of the coating by the prepared roofing.

E-6. *Behavior at 0° C. (32° F.).*—Metal test pieces from E-5, when bent quickly over a 1 inch mandrel immediately after exposure to a temperature of 0° C. (32° F.) for 1 hour shall show no cracking of the coating and no separation of metal and coating.

F. METHODS OF SAMPLING AND TESTING

Deliveries will, in general, be sampled and tested by the following methods, but the purchaser reserves the right to use any additional information to ascertain whether the material meets the specification.

F-1. *Sampling.*—From each shipment the inspector shall take at random a number of packages equivalent to the one half cube root of the total number of packages in the lot. If this proved to be a fractional number, it shall be expressed as the next higher whole number. The contents of each package taken shall be thoroughly stirred until a homogeneous mixture is obtained. One pint of the coating shall then be immediately drawn from each package and transferred to a clean receptable of suitable size, which shall be kept tightly covered except while the coating is being introduced. After all the pint samples have been taken the contents of this container shall be thoroughly stirred and two 1-quart samples of the composite sample shall be transferred to a clean dry container, which shall be immediately stoppered, sealed, and distinctly labeled. One sample shall be transmitted to the testing laboratory and the other shall be retained for check analysis in case of dispute.

F-2. *Laboratory examination.*

F-2a. *Condition in container.*—Open the container when received at the laboratory and examine its contents.

F-2b. *Nonvolatile matter.*—Weigh about 10 grams of the sample in a weighed flat-bottom metal dish about 8 cm in diameter, or in a friction-top can plug. Heat the dish with its contents in an oven at 105° to 110° C. (221° to 230° F.) for 24 hours, cool, and weigh. From the weight of the residue left in the dish and from the weight of the original sample taken compute the percentage of nonvolatile matter.

F-2c. *Mineral filler.*—Weigh from 2 to 5 grams of the coating and transfer to a small beaker. Add 25 to 50 ml of carbon bisulphide, stir until thoroughly mixed, and allow to stand for 15 minutes. Filter the solution through a weighed Gooch crucible prepared with a mat of asbestos, paper pulp, or a paper disc using suction if necessary to aid filtration. Wash the residue in the crucible with cold carbon bisulphide until the washings are colorless, dry the crucible with its contents in air at room temperature until the odor of carbon bisulphide has almost disappeared, and then for 1 hour in an oven at 105° to 110° C. (221° to 230° F.), cool, and weigh. From the weight of the insoluble matter in the crucible and the weight of the sample taken compute the percentage of the matter insoluble in carbon bisulphide and consider it as mineral filler. Ignite a weighed portion of the insoluble matter to a red heat for one-half hour, cool, and weigh.

F-2d. *Behavior at 60° C. 140° (F.).*—Brush the coating at a rate of approximately 1 gallon to 50 square feet on smooth-surfaced asphalt prepared roofing (heavy weight), and clean steel. The steel plates should be about 0.4 mm (0.016 inch) thick. A convenient size for the test pieces is 10 by 15 cm (4 by 6 inches). Note whether the cement spreads easily at room temperature without drawing or pulling. Embed a thread in each coating extending across the coating at a distance of not more than 5 cm (2 inches) from the top of the test piece and measure its distance from the top. After exposing these test pieces for 60 minutes in a horizontal position in a well-ventilated room at a temperature not below 15° C. (59° F.), but not in the direct rays of the sun, suspend them vertically for 5 hours in an oven at 60° C. (140° F.). Remove the test pieces from the oven after 5 hours' heating and examine for blistering and absorption of the coating and for slipping by comparing the distance of the thread from the top of the test piece with its distance at the beginning of the test.

F-2e. *Behavior at 0° C. (32° F.).*—Cool the metal test pieces from F-2d to room temperature and immerse them in a water bath at 0° C (32° F.) for 1 hour. Immediately after their removal from the bath bend the panels in approximately 2 seconds' time, over a mandrel 1 inch in diameter, with the metal next to the mandrel, dry thoroughly, and examine.

G. PACKAGING, PACKING, AND MARKING

G-1. *Packaging.*—Unless otherwise specified, commercial packages are acceptable under this specification.

G-2. *Packing.*—Unless otherwise specified, the subject commodity shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carriers for safe transportation at the lowest rate to the point of delivery.

G-3. *Marking.*—Each package shall be legibly marked with a description of the contents, the name of the manufacturer, the place of manufacture, the brand (if any), the net weight or net liquid measure of the contents, and some means of identification of "the particular contract on which purchase and shipment were made." Each shipment of the material shall be accompanied by printed directions for its use.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H-1. The following departmental specifications, of the issue in effect on date of invitation for bids, shall respectively form a part of this specification.

Orders for this

Specification for Marking Shipments.

H-1b. *Navy:* Navy Department purchases will be made under the issue in effect on date of invitation for bids of Bureau of Yards and Docks Specification No. 7Y, Roofing, Siding, and Sheet Metal Work; Dampproofing and Membrane Waterproofing, copies of which may be obtained without cost upon application to the Bureau of Yards and Docks, Navy Department, Washington, D.C.

I. NOTES

I-1. Asphalt roof coating is intended for use in the repair and coating of asphalt and metal roofing and for application to concrete, masonry, and steel structures as a dampproofing and protective coating. In the application of this material on concrete and masonry it is advisable to apply a priming coat of asphalt primer.

I-2. While the exact nature of the mineral filler to be used in this coating is not specified, asbestos fiber is considered one of the most suitable mineral fillers.

I-3. It is believed that this specification adequately described the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I-4. This specification governs United States Government purchases of this commodity. Users are requested to retain this specification in permanent file until a revision is issued.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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